LEVEL UP ACADEMY

Class 9 | CBSE / GSEB | Science

ANSWER KEY & MARKING SCHEME

Unit Test: Mixtures, Solutions and Separation

Chapter / Topic: Is Matter Around Us Pure (topic-based)

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SECTION A

Q1. Which of the following is a homogeneous mixture? **[1]**

Answer: (c) Salt solution

Q2. Which of these shows the Tyndall effect? **[1]**

Answer: (d) Milk (a colloid)

Q3. Camphor can be separated from common salt by: **[1]**

Answer: (b) sublimation

Q4. Which of these is a chemical change? **[1]**

Answer: (a) Rusting of iron

Q5. Brass is: **[1]**

Answer: (b) a homogeneous mixture (alloy)

Q6. **Assertion (A):** A sugar solution does not show the Tyndall effect. **Reason (R):** The particles in a true solution are too small to scatter light. **[1]**

Answer: (a)

SECTION B

Q7. Classify each as an element, a compound or a mixture: water, air, copper, sugar. **[2]**

Answer: Water: compound. Air: mixture. Copper: element. Sugar: compound.

- ½ each

OR, answer: A saturated solution can dissolve no more solute at a given temperature. For most solids, solubility increases as the temperature increases.

- Definition: 1
- Effect of temperature: 1

Q8. Find the mass of glucose in 250 g of a 20% (mass by mass) glucose solution. **[2]**

Answer: Mass of glucose = 20% of 250 g = (20/100) × 250 = 50 g.

- Method: 1
- 50 g: 1

SECTION C

Q9. Suggest a method to separate: **[3]**

(i) Centrifugation (churning). [1]
(ii) Using a magnet. [1]
(iii) Evaporation. [1]

Q10. What is chromatography? Explain how it separates the colours in ink, and give two uses. **[3]**

Answer: Chromatography separates components of a mixture that dissolve in the same solvent. A spot of ink is put near the bottom of a strip of filter paper, and the strip is dipped in water. As the water rises, the colour that dissolves more readily rises faster and separates from the others. Uses (any two): separating colours in a dye or pigments from natural colours; separating drugs from blood.

- Definition: 1
- How it works: 1
- Two uses: 1

OR, answer: By distillation. The mixture is heated in a distillation flask. Acetone has a lower boiling point (56 °C) than water (100 °C), so it evaporates first, is condensed in the condenser, and is collected separately. Water stays in the flask. This works because the two liquids are miscible but have quite different boiling points.

- Distillation: 1
- Process: 1
- Reason (different boiling points): 1

SECTION D

Q11. Answer the following: **[5]**

(i) Particle size: true solution less than 1 nm; colloid 1–100 nm; suspension more than 100 nm. Visibility: particles of a solution and a colloid cannot be seen with the naked eye; those of a suspension can. Stability: solutions and colloids are stable (particles don't settle); in a suspension the particles settle on standing. [3]

- Each point of comparison: 1

(ii) Milk: colloid. Chalk in water: suspension. Salt solution: true solution. Smoke: colloid. [2]

- ½ each

SECTION E

Q12. While making tea, Priya boiled water, added tea leaves, sugar and milk, and then poured the tea through a strainer into a cup. The tea leaves were… **[4]**

(i) Water. [1]

(ii) Residue: tea leaves. Filtrate: the tea in the cup. [1]

(iii) Filtration (straining). Dissolving sugar is a physical change: no new substance forms, and the sugar can be recovered by evaporating the water. [2]

- Filtration: 1
- Physical change with reason: 1